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Research paper

A novel clustering-ensemble learning model for day-ahead photovoltaic power forecasting

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ABSTRACT

Accurate day-ahead photovoltaic power forecasting (PPF) is essential for grid scheduling and transaction planning. However, time series prediction models based on historical meteorological and photovoltaic power data often struggle to capture long-term dependencies. Existing research typically employs multivariate regression models to establish mapping relationships between numerical weather prediction (NWP) and photovoltaic power data, yet still exhibits deficiencies in weather pattern recognition and consideration of temporal patterns. Therefore, this paper proposes a novel clustering-ensemble learning model to enhance predictive performance. In the proposed deep time-series clustering algorithm, key features from NWP data are extracted for clustering, addressing the inefficiencies of traditional clustering methods in handling high-dimensional data. A weighted Warp-Euclidean distance is proposed to capture sequence trend homogeneity and spatial similarity, overcoming the limitations of Euclidean distance in time-series data. The ensemble learning model combines the advantages of two gradient boosting tree models to handle nonlinear features and high-dimensional data, while introducing lag features to enhance temporal dynamic learning capability. This approach enhances the stability and mitigates the limitations associated with relying primarily on weather-related features. This study utilizes NWP and measured photovoltaic data from two stations in Hebei Province, China. Results demonstrate superior performance compared to other clustering algorithms and prediction models in day-ahead PPF tasks, achieving R^2 scores of 0.952 and 0.943 on two public photovoltaic datasets, representing improvements of 0.020 and 0.017 over optimal benchmark models. The proposed framework provides theoretical guidance for the stable operation of grid-connected photovoltaic systems.

1. Introduction

The growing exhaustion of fossil fuels and rising energy demand have highlighted the critical importance of developing and utilizing renewable energy (He et al., 2025). Grid-connected power generation from new energy sources, particularly solar energy, is a vital component of the future power system. Weather conditions, seasonal variations, and various other factors affect photovoltaic (PV) power, leading to intermittency, fluctuations, and stochastic behavior. These challenges hinder the effective integration of extensive PV facilities into the grid and their reliable functioning within the electrical network. Consequently, precise day-ahead photovoltaic power forecasting (PPF) is significant for both the energy sector and power systems. It can effectively minimize the unpredictability of input energy, enhance system stability, and increase the penetration level of PV systems. It has emerged as a fundamental technology essential for enhancing operational efficiency and dispatch, minimizing reserve capacity storage, and promoting the grid's ability to absorb PV power generation (Tao et al., 2024).

PV output power is closely related to meteorological factors. The high uncertainty of meteorological conditions, combined with the inherently complex patterns of PV generation, makes day-ahead PPF particularly challenging (Li et al., 2023). Time-series forecasting models based on historical data often struggle to capture long-term dependencies, limiting their effectiveness in this context. Multivariate regression methods are frequently adopted to establish mapping relationships between numerical weather prediction (NWP) and PV generation data to improve prediction accuracy. However, existing research faces the

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following challenges: First, PV output power responds significantly differently to various meteorological conditions, making it difficult for a single model to adapt to diverse climate scenarios. Second, NWP data typically exhibits complex, high-dimensional, and nonlinear characteristics. Directly using such data may lead to redundancy and hinder the extraction of key features. Finally, current regression models primarily focus on the direct relationship between NWP data and PV output power, with insufficient attention to temporal dynamic characteristics.

To address these issues, this paper proposes a clustering-ensemble learning model based on NWP and PV output power data from two stations. The model first employs a deep time-series clustering algorithm to extract key features from high-dimensional NWP data, achieving fine-grained division of different meteorological patterns. Subsequently, it integrates an improved ensemble learning model to enhance modeling capabilities for nonlinear and high-dimensional temporal data while effectively adapting to data non-uniformity, thereby improving overall prediction accuracy. The main contributions are summarized as follows:

(1) A novel deep time-series clustering algorithm, termed the convolutional weighted K-medoids clustering (CWK-Medoids), is proposed. Key features from NWP data are extracted for clustering by introducing the convolutional autoencoder (CAE), addressing the computational intensity and inefficiencies of traditional clustering methods in handling high-dimensional data. Additionally, a weighted Warp-Euclidean distance (WWED) is constructed to account for sequence trend homogeneity and spatial similarity, overcoming the inadequacy of Euclidean distance in capturing time-series features.

(2) An improved ensemble learning model, termed the lag features-based boosting network (LagBoostNet), is introduced. Two gradient boosting tree models are utilized to handle nonlinear features and high-dimensional data, overcoming the instability often associated with single models. By introducing lag features, the model converts historical power data into effective model-learnable features, enhancing temporal dependency modeling. This addresses the limitation in existing regression models, which typically focus on weather correlations while falling short in capturing temporal dynamics.

The content is organized as follows: Section 2 offers a comprehensive overview of related work. Section 3 outlines the methodology, model parameters, and evaluation metrics. Section 4 presents the dataset and compares the suggested approach with three clustering methods and eight forecasting models. The approach is interpreted, and ablation experiments are performed to assess the impact of individual modules on the model. Section 5 offers a conclusion and proposes potential avenues for further study.

2. Related work

Day-ahead PPF mainly focuses on accurately forecasting PV output power one day in advance (Perera et al., 2022). Efficient transaction management and plant scheduling rely on an accurate day-ahead PPF model. Depending on the scheduling requirements of the PV plant, forecasts are categorized into 1-h and 15-min intervals.

PPF techniques are commonly categorized into physical modeling methods and data-driven algorithms (Liu et al., 2025). Physical models simulate PV system operations using weather inputs (e.g., solar irradiance, ambient temperature) to predict output power (Mayer and Gróf, 2021). For instance, a two-step physical PPF was implemented by converting localized NWP radiation data into power forecasts (Wang et al., 2024b). However, these models face limitations: (1) Limited applicability to specific regions and PV plant types, often requiring specialized equipment and strict environmental conditions; (2) High computational complexity due to extensive data processing for parameter estimation; (3) Dependency on rigid parameters/assumptions that may not reflect real-world variability; (4) Prediction accuracy may gradually decline over time as PV system operating parameters potentially change.

Data-driven algorithms leverage datasets (e.g., weather data, ground and satellite measurements, NWP, and historical PV generation) to model the relationship between influencing factors and power output (Gu et al., 2021). With advances in data quality and technology, these methods have become widely used for day-ahead PPF (Qu et al., 2021). Recent studies have developed time-series forecasting models using historical meteorological and PV power data. An innovative transformer-based model designed to forecast PV generation for 24, 36, and 40 h ahead, was introduced (Tao et al., 2024). A hybrid model that incorporated a transformer and an optimized approach for intraday and day-ahead PPF was developed (Wang and Ma, 2024). However, such methods struggle to capture long-term dependencies in day-ahead forecasting while facing high computational costs and feature extraction challenges. Consequently, some studies have shifted focus to direct regression modeling between weather forecasts and PV generation data to enhance prediction accuracy. The proposed model leveraged NWP, ground irradiance, and PV system data to address day-ahead PPF under imperfect measurements (Massidda et al., 2024). An innovative dual-phase hybrid framework that combined NWP and past feature extraction techniques for 15-min interval PPF was introduced, demonstrating high accuracy and adaptability (Li et al., 2023). Ensemble algorithms are highly effective in real-world applications, often improving the performance of individual base models (Zheng et al., 2025). For instance, an enhanced stacked integration algorithm, which employed dual deep learning algorithms as base models, was introduced for PPF (Khan et al., 2022). Such algorithms can enhance the accuracy and stability of day-ahead forecasting to some extent. However, it mostly concentrates on basic-level model integration, primarily considers the correlation between PV output power and meteorological factors, while insufficiently considers its temporal characteristics. Therefore, it is necessary to explore hybrid and ensemble model architectures, combine different models to leverage their respective advantages, and introduce feature engineering modules to mine latent patterns in time series.

Meanwhile, significant differences in PV output power under varying weather conditions necessitate the analysis of similar target day sets by modeling each category separately (Dou et al., 2024). Clustering partitions the data into multiple scenarios, each representing a subset of the original temporal dataset. Targeted simulation within each cluster can better capture inter-scenario differences and enables the model to adapt to data non-uniformity. This approach more accurately fits the data within each scenario, reduces prediction errors and model complexity, and ultimately enhances forecasting performance (Tang et al., 2024).

Clustering techniques are typically categorized into conventional clustering and deep clustering algorithms. The former is well suited for low-dimensional data (Chatterjee and Choudhury, 2025), which is commonly encountered in PPF applications. Fuzzy c-means clustering (FCM) was applied to NWP and historical PV data, in combination with optimization techniques, to perform day-ahead PPF (Gu et al., 2021). K-means++ was used to classify the dataset into three distinct weather conditions, based on features selected through correlation analysis. These clusters were then used in combination with advanced techniques for PPF (Li et al., 2024). The aforementioned studies indicate that clustering preprocessing of data can enhance prediction accuracy. However, conventional clustering algorithms are heavily influenced by the underlying characteristics of the data. When applied to high-dimensional scenarios, they often suffer from increased computational complexity, making it difficult to extract meaningful patterns. In PPF applications, clustering based on multiple weather features often involves correlation analysis to remove highly correlated variables. However, this process may inadvertently eliminate important features, resulting in suboptimal clustering outcomes. Deep clustering refers to a class of algorithms that utilize the nonlinear representation and feature extraction capabilities of deep learning for clustering tasks. Many existing methods are based on the autoencoder (AE) framework. For instance, a deep non-negative matrix factorization (NMF)

Table 1
Day-ahead PPF methods.

References	Main structure	Data
Tao et al. (2024)	Data augmentation methods for physical modeling and Transformer variants	Metadata, NWP, and local measurement data
Wang and Ma (2024)	Dynamic mean preprocessing algorithm, improved whale variational mode decomposition, and context-embedded causal convolution Transformer	PV power generation and meteorological data
Li et al. (2024)	K-means++, variational mode decomposition (VMD), and bidirectional long short-term memory (BiLSTM)	
Massidda et al. (2024)	Conformal quantile regression	PV power generation, NWP, and satellite monitoring data
Li et al. (2023)	NWP information mapping and historical feature extraction methods	PV power generation and NWP data
Khan et al. (2022)	Artificial neural network (ANN), long short-term memory (LSTM), and extreme gradient boosting (XGBoost)	
Gu et al. (2021)	FCM, whale optimization algorithm (WOA), least squares support vector machine (LSSVM), and nonparametric kernel density estimation (KDE)	

algorithm inspired by AE achieved superior performance on eight datasets (Wang et al., 2024a). A deep clustering network leveraging multi-representation AE and auxiliary techniques was also proposed (Wen et al., 2023). Additionally, AE-based models combined with other dimensionality reduction techniques have been developed to address complex clustering problems (Chatterjee and Choudhury, 2025). Building on these developments, deep clustering has been extended to sequential data analysis. A method integrating topological information with time series clustering was introduced, effectively capturing temporal patterns to produce cluster-specific representations (Lee et al., 2024). In conclusion, deep clustering can seamlessly integrate clustering objectives with the expressive capabilities of deep learning, offering clear advantages over traditional methods. It is particularly suitable for handling complex, high-dimensional, and nonlinear weather data, as it can automatically extract more representative features. However, it has seldom been applied in the field of PPF. Moreover, existing deep clustering approaches struggle to fully capture time-dependent characteristics. The time-shift effect, in which similar patterns are misaligned temporally, can lead to the incorrect grouping of analogous curves into different clusters. The main existing approaches for day-ahead PPF are summarized, as shown in Table 1.

3. Methodology

The overall structure of the suggested approach is first outlined. Subsequently, the principles of CWK-Medoids and LagBoostNet are described. Section 3.4 outlines the model parameters and the evaluation criteria.

3.1. Forecasting procedure of the proposed method

We propose a day-ahead PPF approach that integrates a novel deep time-series clustering algorithm, CWK-Medoids, with an improved ensemble learning model, LagBoostNet. After clustering, the approach models NWP data and PV output power for each scenario separately. CWK-Medoids employs a CAE to extract low-dimensional features from high-dimensional NWP inputs, and evaluates sequence similarity jointly in temporal and spatial dimensions. It introduces a new similarity measure, WWED, which combines dynamic time warping (DTW) and Euclidean distance to capture both trend homogeneity and spatial coherence. LagBoostNet, on the other hand, fuses CatBoost and LightGBM to leverage their strengths in modeling nonlinear and high-dimensional features. By incorporating lag features, it captures autocorrelation and temporal dependencies, enabling the extraction of meaningful patterns from time series fluctuations. This combined framework effectively adapts to data nonuniformity, ensures accurate scenario-specific modeling, and enhances both prediction accuracy and generalization for day-ahead PPF. Fig. 1 illustrates the overall structure of the proposed approach.

Importantly, our PPF task does not follow the conventional time series paradigm that predicts power on day $(D+f)$ using historical data up to day D. Instead, we adopt a day-ahead framework: at time $D-1$, we obtain 15-min interval NWP data for day D, which are then mapped by a regression model to the corresponding PV power output on that day. This ensures the prediction is completed before day D begins, relying solely on forecast inputs from $D-1$, and avoids any access to actual PV observations from day D or beyond. Accordingly, both clustering and prediction strictly utilize data available before day D, eliminating the risk of future information leakage. Additionally, CWK-Medoids clustering serves as an unsupervised preprocessing step that is fully independent of the subsequent prediction. (1) Clustering phase: The entire dataset is partitioned into mutually exclusive meteorological scenarios using only NWP features, without any target variable involvement. Clustering performance is assessed via two intrinsic metrics on the complete dataset (Section 3.4.2(1)). (2) Prediction phase: Independent prediction models are constructed for each cluster. Within each subset, data is partitioned into training set (55%), validation set (10%), and test set (35%), with LagBoostNet training neither invoking clustering model parameters nor outputs. Prediction accuracy is solely evaluated on each cluster's test set through four metrics (Section 3.4.2(2)). This design ensures that: (i) clustering remains fully unsupervised, thereby eliminating the risk of label information leakage; (ii) test sets are never exposed during model training, and no information is shared across clusters; and (iii) clustering and prediction workflows remain independent, with their effectiveness and performance assessed separately. Fig. 2 illustrates the workflow of the proposed approach.

3.2. Deep time-series clustering model: CWK-medoids

CWK-Medoids is suggested for clustering data from PV systems while considering multiple weather factors. Multiple daily NWP series data are used as input. The elbow approach reveals that the optimal cluster count is found to be six. It primarily consists of the following structures:

3.2.1. CAE

The autoencoder (AE) typically comprises an encoder and a decoder. The former compresses the input into a compact feature representation, while the latter reconstructs it to generate a result that closely resembles the original input (Zheng et al., 2024). For an AE with a single hidden layer, in light of the input $\eta \in \mathbb{R}^d$, the encoded data $z \in \mathbb{R}^{d'}$ of $d' \ll d$ becomes:

$$z = \sigma(W_{encoder}\eta + b_{encoder}) \tag{1}$$

where σ represents the nonlinear activation function. The hidden layer's weights are denoted by $W_{encoder} \in \mathbb{R}^{d' \times d}$, and its biases are indicated by $b_{encoder} \in \mathbb{R}^{d'}$. The output $\eta' \in \mathbb{R}^d$ can be derived from the following equation:

$$\eta' = \sigma(W_{decoder}z + b_{decoder}) \tag{2}$$

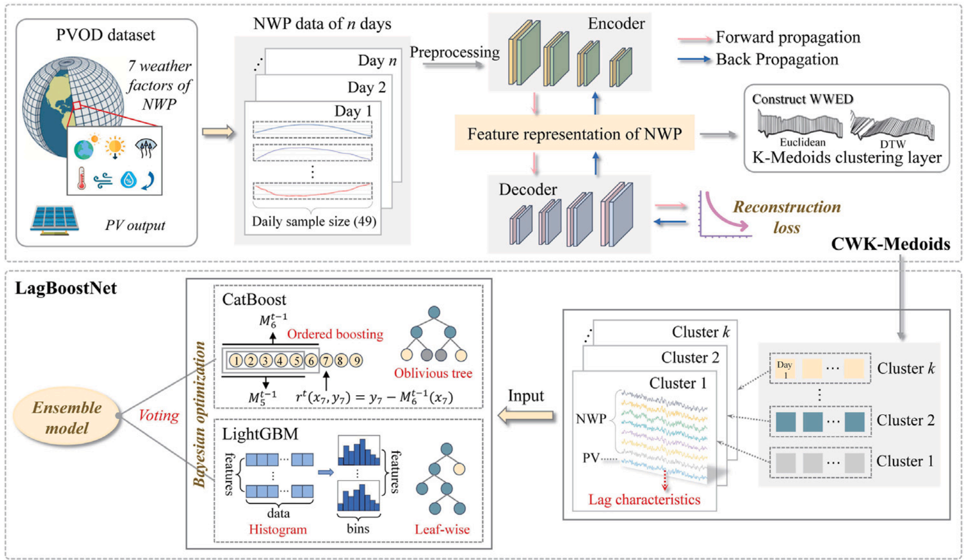

Fig. 1. Overall structure of the suggested approach.

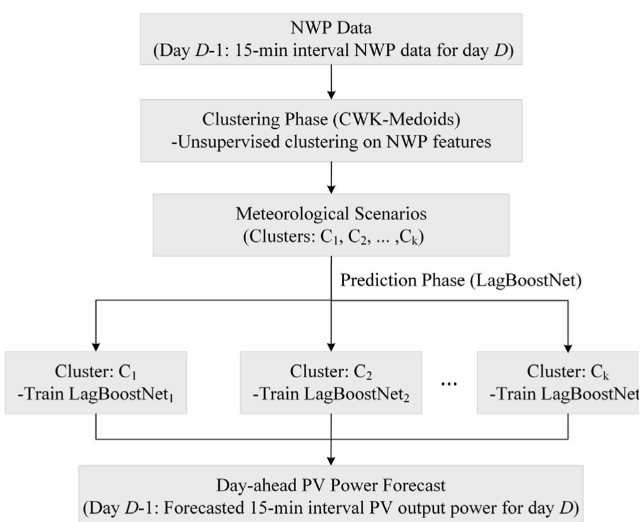

Fig. 2. Flowchart of the proposed approach.

During training, the AE updates its weights through a backpropagation algorithm to minimize the reconstruction error. The objective function typically employs mean squared error (MSE), with the reconstruction loss L_c defined as follows:

$$L_c = \frac{1}{n}\|\eta - \eta'\|_2^2 \tag{3}$$

CAE is a variant of AE, where the primary difference lies in the incorporation of convolutional and pooling layers, as opposed to fully connected layers. This enables the retention of spatial features in the original data while more efficiently extracting its depth features. In the convolutional layer, feature maps are generated by performing convolutional operations on the input (or past feature maps) and filters. Using two-dimensional convolution, the process in the CAE convolutional layer is performed by substituting matrices A and B with the associated filters and feature maps:

$$A(\lambda,\mu) * B(\lambda,\mu) = \sum_{T_1=-\infty}^{\infty}\sum_{T_2=-\infty}^{\infty} A(\lambda - T_1, \mu - T_2)B(T_1, T_2) \tag{4}$$

where * corresponds to the convolution operator. λ refers to the row index, whereas μ signifies the column index. Designate the kth feature map of the lth layer as $F_l^k \in \mathbb{R}^{x_l \times y_l}$, the relevant filter be $W_{k'}^k \in \mathbb{R}^{x_l' \times y_l'}$, and convolve it via the k'th feature map originating from the preceding layer $F_{l-1}^{k'}$, where x_l, y_l (x_l', y_l') refers to the dimensions of the feature map (filter) in the lth layer. Subsequently, F_l^k can be derived through the following process:

$$F_l^k(\lambda,\mu) = \sigma\left(\sum_{k'=1}^{d_{l-1}} W_{k'}^k(\lambda,\mu) * F_{l-1}^{k'}(\lambda,\mu) + b_k\right) \tag{5}$$

where d_{l-1} stands for the depth of layer l-1. σ employs the ReLU activation function (Ryu et al., 2019).

The pooling layer progressively decreases the feature map's dimensions by iteratively selecting values from predefined spatial regions. Maximum pooling is employed here:

$$h_{pool} = \max\{h\} \tag{6}$$

where h_{pool} represents the result generated by the pooling layer, indicating the encoded representation after pooling.

The encoder utilizes convolution and pooling layers for feature extraction, whereas the decoder utilizes deconvolution and upsampling layers to reconstruct the input. The deconvolution layer serves to expand the feature map and restore the spatial structure of the data. The upsampling layer increases the data size to facilitate its reconstruction.

$$h_{upsample} = Interpolate(h_{pool}) \tag{7}$$

where $h_{upsample}$ denotes the output of the upsampling layer. $Interpolate(\cdot)$ represents the interpolation operation, with nearest neighbor interpolation applied in this case.

3.2.2. Clustering layer

The clustering algorithm utilizes a similarity metric to partition the dataset into multiple clusters to minimize the extent of similarity among data points within a single cluster. In contrast, the dissimilarity between data objects in different clusters is maximized. K-means clustering is sensitive to noise points and outliers, making it prone to converging to local optima. K-Medoids clustering selects the sample most similar to others in each cluster as the cluster center, addressing the limitations of K-Means clustering (Pei et al., 2023). Hence, the clustering layer is implemented using K-Medoids.

Existing deep clustering algorithms predominantly utilize Euclidean distance as a similarity measure, although Euclidean distance fails to capture the morphological characteristics of the curve accurately. This paper thoroughly examines the trend homogeneity and spatial similarity of daily NWP series data, combining Euclidean distance and DTW distance to construct WWED for calculating the similarity of time series (Formula (8)–(11)).

Given two temporal data, $\mathrm{H} = [\eta_1, \eta_2, \ldots, \eta_n]$ and $\Theta = [\theta_1, \theta_2, \ldots, \theta_m]$, $n, m \in N$, the Euclidean distance between H and Θ is expressed in the following manner:

$$Dist_{ED}(\mathrm{H}, \Theta) = \sqrt{\sum_{t=1}^{n} (\eta_t - \theta_t)^2} \tag{8}$$

Unlike Euclidean distance, which computes the distance between the matching values of two temporal data at the same time point, DTW distance computes the morphological similarity by aligning and mapping the time series by warping the time axis. For H and Θ, define $D \in R^{n \times m}$ as the distance matrix between them, and its element $d_{\lambda\mu}$ is:

$$d_{\lambda\mu} = (\eta_\lambda - \theta_\mu)^2, \lambda \in [1, n], \mu \in [1, m] \tag{9}$$

Let $W = [\omega_1, \omega_2, \ldots, \omega_l]$ denote the DTW path vector of time series H and Θ, where $l \in N, \omega_p = d_{\lambda\mu}, p \in [1, l]$ represents the time point η_λ corresponding to θ_μ. Vector W must satisfy the following conditions: (1) boundary constraint, i.e., $\omega_1 = d_{11}$ and $\omega_l = d_{nm}$; (2) path length constraint, i.e., $\max(m, n) \le l \le m + n - 1$; (3) path continuity constraint, i.e., if $\omega_p = d_{\lambda\mu}, \omega_{p+1} = d_{\lambda'\mu'}$, there should be $0 \le \lambda' - \lambda \le 1$ and $0 \le \mu' - \mu \le 1$. The original DTW algorithm may suffer from an ill-conditioned data alignment issue, where a single point in one time series continuously corresponds to multiple points in another time series. As a result, the two curves obtained through long-distance time-axis translation will yield smaller morphological distance values. Thus, the Sakoe-Chiba constraint is incorporated, building upon the aforementioned constraints to restrict the DTW path. The Sakoe-Chiba constraint is defined as follows: if $\omega_p = d_{\lambda\mu}$, then $|\mu - \lambda| \le \delta, \delta \in [\min(m, n), \max(m, n)]$. Here, δ represents the maximum path bending length. This constraint effectively limits the maximum number of consecutive correspondences between single points in time series, ensuring that only time series with high overlap after short-time translation can achieve a smaller morphological distance. There are multiple paths, and the DTW distance is defined as the distance along the path that minimizes the total distance.

$$Dist_{DTW}(\mathrm{H}, \Theta) = \arg\min_{W}\left(\sqrt{\sum_{i=1}^{l} \omega_\lambda}\right) \tag{10}$$

The Euclidean distance and the DTW distance are combined to construct the WWED:

$$Dist_{WWED} = \alpha \cdot Dist_{ED}(\mathrm{H}, \Theta) + (1 - \alpha) \cdot Dist_{DTW}(\mathrm{H}, \Theta) \tag{11}$$

where α refers to the weight associated with the Euclidean distance, and $1 - \alpha$ corresponds to the weight of the DTW distance. To identify the optimal weight combination, we systematically implemented the grid search method for parameter optimization within a predefined range, ultimately assigning α a value of 0.5.

3.3. An improved ensemble learning model: LagBoostNet

LagBoostNet is an improved ensemble learning model based on lag features. The weighted voting strategy integrates the two gradient boosting tree models, CatBoost and LightGBM, with hyperparameters automatically adjusted through Bayesian optimization. The historical power pattern and its trend over time can be effectively learned by introducing lag features.

3.3.1. CatBoost

CatBoost is built upon the gradient-boosting decision tree (GBDT). It efficiently handles categorical features and builds an unbiased gradient tree, enhancing generalization and prediction accuracy (Prokhorenkova et al., 2018).

Categorical features are discrete variables, typically represented as strings. They are not suitable for direct use as input and must undergo processing by shuffling the order of a numeric value, $D = \{(\eta_\lambda, \theta_\lambda)\}_{\lambda=1,\ldots,n}$. The arrangement after shuffling is denoted as $\sigma = (\sigma_1, \ldots, \sigma_n)$, and by traversing from σ_1 to σ_n, the categorical feature's attribute is computed from the zth entry.

$$\sigma_{z,k} = \frac{\sum_{\mu}^{z-1} [\eta_{\sigma_{\mu,k}} = \eta_{\sigma_{\lambda,k}}] \cdot \Theta_{\sigma_\mu} + \alpha \cdot z}{\sum_{\mu}^{z-1} [\eta_{\sigma_{\mu,k}} = \eta_{\sigma_{\lambda,k}}] + \alpha} \tag{12}$$

where z symbolizes the prior term, and $\alpha > 0$ symbolizes its weight coefficient. This helps mitigate noise within categorical features.

CatBoost employs Ordered Boosting to modify GBDT, mitigating the impact of gradient estimation bias. Specifically, a permutation σ of [1, n] is randomly generated, and the initial samples are arranged utilizing σ. It is initialized with n diverse models $M_1, M_2, \ldots, M_n$, where each M_λ utilizes exclusively the first λ specimens of the stochastic arrangement. At each iteration, the unbiased gradient estimate for the μth sample is computed with model $M_{\mu-1}$ (Yu et al., 2024). Additionally, CatBoost employs the oblivious tree as its base predictor.

3.3.2. LightGBM

LightGBM is a Boosting-based ensemble approach that builds a robust regression tree by aggregating multiple weak ones (Ke et al., 2017):

$$F(\eta) = \sum_{k=1}^{K} f_k(\eta) \tag{13}$$

where $F(\eta)$ represents the sample prediction, and $f_k(\eta)$ signifies the result produced by the weak regression tree.

LightGBM incorporates various enhancements derived from GBDT. First, a histogram-based decision tree algorithm partitions continuous variables into M discrete intervals. It creates a histogram from the partitioned elements, continuously processing the input until the decision tree is fully developed (Lu et al., 2024). Second, a leaf-wise leaf growth strategy is adopted to boost learning and mitigate overfitting through depth restriction. Furthermore, the gradient-based one-side sampling (GOSS) approach is utilized to reduce the sample size by eliminating most samples with minimal gradients and to assess the information gain solely for the remaining subset of samples. Finally, the exclusive feature bundling (EFB) approach combines mutually exclusive features with the aim of feature reduction. It utilizes a histogram-based approach to create synthetic features, effectively decreasing the data's dimensionality. These techniques boost prediction accuracy, speed up training for large datasets, and minimize memory usage (Song and Chen, 2024).

3.3.3. Lag features

Lag features are added to transform historical power into features that the model can learn. Let $P(t)$ denote the PV power at time t. Lag features are constructed as follows:

$$P_{lag_k}(t) = P(t - k) \tag{14}$$

where k represents the time step of the lag. In this study, k = 49, 98, 147 was set to introduce three lag features, representing the power values simultaneously on the previous day, two days prior, and three days prior, respectively.

A logarithmic transformation is applied to the lag features to decrease the feature variance, thereby improving the model's fitness to the data:

$$P^{\log}_{lag_k}(t) = \log(P_{lag_k}(t)) \tag{15}$$

Table 2
Hyperparameters of LagBoostNet at the 1st PV station.

Submodel	Symbol	Range	Parameter values for each clustering scenario					
			0	1	2	3	4	5
LightGBM	lambda_l1	(1e−3, 10.0)	1.791	0.001	0.043	0.002	0.001	0.002
	lambda_l2	(1e−3, 10.0)	0.003	1.627	0.022	0.176	3.715	0.588
	lr	(1e−2, 1e−1)	0.022	0.012	0.013	0.030	0.014	0.010
	max_tr_depth	(4, 8)	4	7	6	8	5	8
	num_leaves	(16, 256)	26	214	158	150	228	225
	feat_ratio_tree	(0.4, 1.0)	0.876	0.490	0.417	0.422	0.402	0.455
	feat_ratio_node	(0.4, 1.0)	0.446	0.401	0.427	0.400	0.485	0.476
	bagging_fraction	(0.4, 1.0)	0.773	0.682	0.667	0.448	0.517	0.999
	bagging_freq	(1, 7)	1	1	4	2	5	1
	min_data_in_leaf	(5, 100)	5	84	41	61	14	99
	scale_pos_weight	(0.8, 4.0)	3.750	3.143	1.765	2.965	3.368	1.334
CatBoost	lr	(1e−2, 1e−1)	0.022	0.012	0.017	0.023	0.015	0.013
	max_tr_depth	(4, 8)	8	4	4	5	4	8
	lambda_l2	(1e−3, 10.0)	3.916	3.022	7.311	0.003	0.823	0.002
	feat_ratio_level	(0.4, 1.0)	0.963	0.694	0.971	0.451	0.666	0.482
	min_data_in_leaf	(5, 100)	54	5	82	61	25	43

Table 3
Hyperparameters of LagBoostNet at the 2nd PV station.

Submodel	Symbol	Range	Parameter values for each clustering scenario					
			0	1	2	3	4	5
LightGBM	lambda_l1	(1e−3, 10.0)	0.008	0.230	0.004	1.965	0.035	0.006
	lambda_l2	(1e−3, 10.0)	1.712	0.002	0.299	0.019	0.061	0.003
	lr	(1e−2, 1e−1)	0.067	0.026	0.020	0.020	0.020	0.010
	max_tr_depth	(4, 8)	5	5	4	8	4	7
	num_leaves	(16, 256)	84	108	111	74	65	199
	feat_ratio_tree	(0.4, 1.0)	0.580	0.823	0.400	0.975	0.541	0.495
	feat_ratio_node	(0.4, 1.0)	0.939	0.426	0.525	0.467	0.437	0.537
	bagging_fraction	(0.4, 1.0)	0.536	0.538	0.930	0.734	0.988	0.986
	bagging_freq	(1, 7)	3	7	1	5	5	5
	min_data_in_leaf	(5, 100)	25	98	18	39	29	31
	scale_pos_weight	(0.8, 4.0)	2.756	1.937	2.167	2.395	2.235	2.218
CatBoost	lr	(1e−2, 1e−1)	0.044	0.033	0.036	0.020	0.023	0.011
	max_tr_depth	(4, 8)	4	4	7	6	4	5
	lambda_l2	(1e−3, 10.0)	0.002	0.019	0.003	1.169	4.832	0.150
	feat_ratio_level	(0.4, 1.0)	0.890	0.618	0.698	0.628	0.474	0.630
	min_data_in_leaf	(5, 100)	50	97	59	9	53	20

3.4. Model parameters and evaluation metrics

3.4.1. Model parameters

For CWK-Medoids, the model is trained using 500 epochs, a learning rate of 0.001, the MSE reconstruction loss function, and the Adam optimizer. For LagBoostNet, the objective function of LightGBM is the L1 loss function, while that of CatBoost is the mean absolute error (MAE), with 200 iterations. Optuna is employed for Bayesian optimization to identify the optimal hyperparameter combinations for LagBoostNet by maximizing the R^2 scores of cross-validation. The range of each hyperparameter is defined within values consistent with its intended meaning. The hyperparameters of LightGBM include the L1 and L2 regularization coefficients, learning rate, maximum tree depth, maximum number of tree leaves, feature sampling ratios for each tree and node, data sampling ratio, frequency of data sampling, minimum samples per leaf node, and weight adjustment for category imbalance. The hyperparameters of CatBoost include the learning rate, maximum tree depth, L2 regularization coefficient, feature sampling ratio of each level node, and the minimum samples per leaf node. The hyperparameters of the LagBoostNet model for the two PV sites are provided in Tables 2 and 3.

3.4.2. Evaluation metrics

(1) Evaluation of clustering effect

The clustering effect is evaluated through the Calinski–Harabasz (CH) index and shape-based dissimilarity (SBD). These metrics assess the compactness, separation, and curve shape similarity of the clustering results. A larger CH value and an SBD value closer to 0 indicate better clustering results. The details are described as follows:

A. CH

The CH index must account for both the intraclass variance W and the interclass variance B. W represents the total of the squared deviations between all intracluster observations and the center of their respective cluster:

$$W = \sum_{\lambda=1}^{k} \sum_{\mu=1}^{n_\lambda} (g_{\lambda\mu} - o_\lambda)^2 \tag{16}$$

where n_λ, $g_{\lambda\mu}$, and o_λ refer to the total number of observations, the μth observation, and the centroid of the λth group, respectively.

The interclass variance B is employed to quantify the dispersion of data points between clusters:

$$B = \sum_{\lambda=1}^{k} n_\lambda (\bar{g} - o_\lambda)^2 \tag{17}$$

where $\bar{g}$ signifies the mean of the entire dataset. The CH index is derived through the following formula:

$$CH = \frac{B}{W} \frac{n-k}{k-1} \tag{18}$$

B. SBD

Given that traditional clustering methods primarily focus on spatial similarity between sequences, the algorithm proposed in this study additionally considers their trend homogeneity. Therefore, SBD, which can effectively measure trend similarity between sequences, is selected to evaluate the proposed algorithm's improvement in trend feature extraction. When the sequence θ is shifted from its initial position with a step size of 1, the data at the original position of sequence θ is replaced with 0 in the portion of θ where the horizontal coordinates do not coincide with those of sequence η. Thus, the cross-correlation sequence between η and θ is denoted as $CC_{\omega}(\eta,\theta) = (c_1,\ldots,c_{\omega})$. The duration of this sequence is ($2T$-1). The following represents the definition (Paparrizos and Gravano, 2017):

$$\begin{cases} CC_{\omega}(\eta,\theta) = R_s(\eta,\theta), \omega \in \{1,2,\ldots,2T-1\} \\ s = \omega - T \end{cases} \tag{19}$$

$$R_s(\eta,\theta) = \begin{Bmatrix} \sum_{l=1}^{T-s} (\eta_{l+s} \cdot \theta_l), s \geq 0 \\ R_{-s}(\theta,\eta), s < 0 \end{Bmatrix} \tag{20}$$

where s denotes the moving step of sequence θ, $s \in [-T,T]$. ω refers to the position subscript of sequence $CC_{\omega}(\eta,\theta)$. $R_s(\eta,\theta)$ represents the sum of dot products from the overlapping parts between the two sequences at each step of sequence θ. The position $\omega_{\max}$ corresponding to the maximum mutual correlation value can be derived from sequence $CC_{\omega}(\eta,\theta)$. The coefficient normalization technique is utilized to transform the sequences, enhancing the comparability of the mutual correlation sequence data. This ensures the zero-lag autocorrelation coefficient equals 1. The coefficient normalization formula is given by:

$$NCC_c(\eta,\theta) = \frac{CC_{\omega}(\eta,\theta)}{\sqrt{R_0(\eta,\eta)\cdot R_0(\theta,\theta)}} \tag{21}$$

where $NCC_c(\eta,\theta)$ represents the coefficient-normalized cross-correlation sequence, which has a value range of $[-1, 1]$. $R_0(\eta,\eta)$ and $R_0(\theta,\theta)$ signify the autocorrelations for the sequences η and θ, in that order. When $NCC_c(\eta,\theta)$ equals 1 or -1, the two sequences are fully positively or negatively correlated. Therefore, a larger value of $NCC_c(\eta,\theta)$ indicates greater similarity between the two sequences. To simplify the calculation, the similarity distance between the morphological trends of sequences η and θ is expressed as SBD:

$$SBD(\eta,\theta) = 1 - \max(NCC_c(\eta,\theta)) \tag{22}$$

where the SBD value range is [0, 2], with 0 indicating that the morphological trends of sequences η and θ are fully similar, and 2 indicating that they are entirely opposite. The SBD of the clustering algorithm represents the average value of SBD across each clustering scenario.

(2) Evaluation of prediction accuracy

The coefficient of determination (R^2) evaluates the goodness of fit, root mean squared error ($RMSE$) measures prediction error, mean absolute error (MAE) assesses the average magnitude of errors, and weighted absolute percentage error ($WAPE$) calculates the relative prediction error. Higher R^2 values and lower values of the other three indicators signify greater accuracy, as detailed below (Brester et al., 2023):

$$R^2 = 1 - \frac{\sum_{\lambda=1}^{n} (\theta_{\lambda} - \hat{\theta}_{\lambda})^2}{\sum_{\lambda=1}^{n} (\theta_{\lambda} - \bar{\theta})^2} \tag{23}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{\lambda=1}^{n} (\theta_{\lambda} - \hat{\theta}_{\lambda})^2} \tag{24}$$

$$MAE = \frac{1}{n}\sum_{\lambda=1}^{n} \left|\theta_{\lambda} - \hat{\theta}_{\lambda}\right| \tag{25}$$

$$WAPE = \frac{\sum_{\lambda=1}^{n} |\theta_{\lambda} - \hat{\theta}_{\lambda}|}{\sum_{\lambda=1}^{n} |\theta_{\lambda}|} \times 100\% \tag{26}$$

where n stands for the sample size, θ_{λ} refers to the actual value, $\hat{\theta}_{\lambda}$ stands for the predicted value, and $\bar{\theta}$ corresponds to the mean of the actual values.

4. Results and discussion

This chapter begins by presenting the dataset employed in this study. The effectiveness of various clustering algorithms is then evaluated, followed by comparing the proposed model's performance with eight forecasting methods for day-ahead PPF. Ultimately, the model interpretation is carried out, followed by ablation experiments to verify the impact of individual modules. All experiments are conducted on Intel(R) Core(TM) i7-13700K (24 CPUs), ~3.4 GHz, NVIDIA GeForce RTX 4070 Ti, with Python 3.10 and PyTorch 2.4.1.

4.1. Data description

The PV power output dataset (PVOD) comprises NWP and local measurement data collected from various PV systems in Hebei Province, China (36.64403–39.5155° N, 113.6419–117.4572° E). The data is collected over the period from July 1, 2018, to June 13, 2019, with a time resolution of 15 min (https://github.com/yaotc/PVODataset). A specific variant of the weather research and forecasting (WRF) model is employed to generate the NWP data. It operates once daily, utilizing a horizontal grid resolution of 4 km to retrieve NWP data from the horizontal plane, covering a forecast period of 28 to 54 h. The NWP data and their corresponding measured PV data from two PV systems are selected for the experiment. The NWP data includes seven weather factors: global irradiance (W/m^2), temperature (°C), atmospheric pressure (hPa), direct irradiance (W/m^2), wind speed (m/s), wind direction (°), relative humidity (%). The measured PV data refers to the PV output of the station (MW). It is noteworthy that PV output power exhibits strong dependence on meteorological factors, with day-ahead forecasting being particularly sensitive to the quality of NWP data. This sensitivity arises from the fact that NWP data, generated by numerical models, may contain systematic errors and suffer from regional adaptability issues. These errors can propagate directly into the inputs of prediction models, potentially compromising predictive performance (Massidda et al., 2024). The dataset used in this study has been previously published and peer-reviewed (Yao et al., 2021), and has also been successfully applied in PV forecasting research (Tao et al., 2024; Yao et al., 2022), confirming its quality and applicability.

During the data preprocessing phase, multiple tasks were performed to enhance data quality: (1) To eliminate interference from abnormal power generation status of PV power stations on model training, raw samples were screened to exclude abnormal daily samples exhibiting missing rates exceeding 30% or prolonged zero power generation during daytime hours (07:00-19:00). Such conditions typically arise from equipment malfunctions or data acquisition anomalies. (2) For missing values in the remaining valid data, K-nearest neighbor (KNN) interpolation was employed to mitigate potential impacts on model performance. (3) Considering numerical scale differences among features may affect distance calculations in clustering algorithms, all meteorological variables were normalized prior to clustering to prevent dominance by variables with larger numerical ranges. (4) Given the physical characteristic of zero nighttime power output in PV systems, only valid generation periods (07:00-19:00 daily) were retained for modeling. This reduced interference from numerous zero values during model training while enhancing modeling efficiency and prediction accuracy. The final sample sizes are 17,052 and 16,856, respectively.

4.2. Verification of clustering effectiveness

CWK-Medoids is compared to three clustering algorithms to assess the effectiveness of the novel deep time-series clustering algorithm. These algorithms include spectral cluster (SC), empty circles-based k-means (ECKM), and density peak clustering (DPC). Specifically, SC (Zheng et al., 2023) is a clustering approach grounded in graph theory, which is particularly suitable for addressing clustering problems in complex data. ECKM (Biswas et al., 2023) is an enhanced K-means

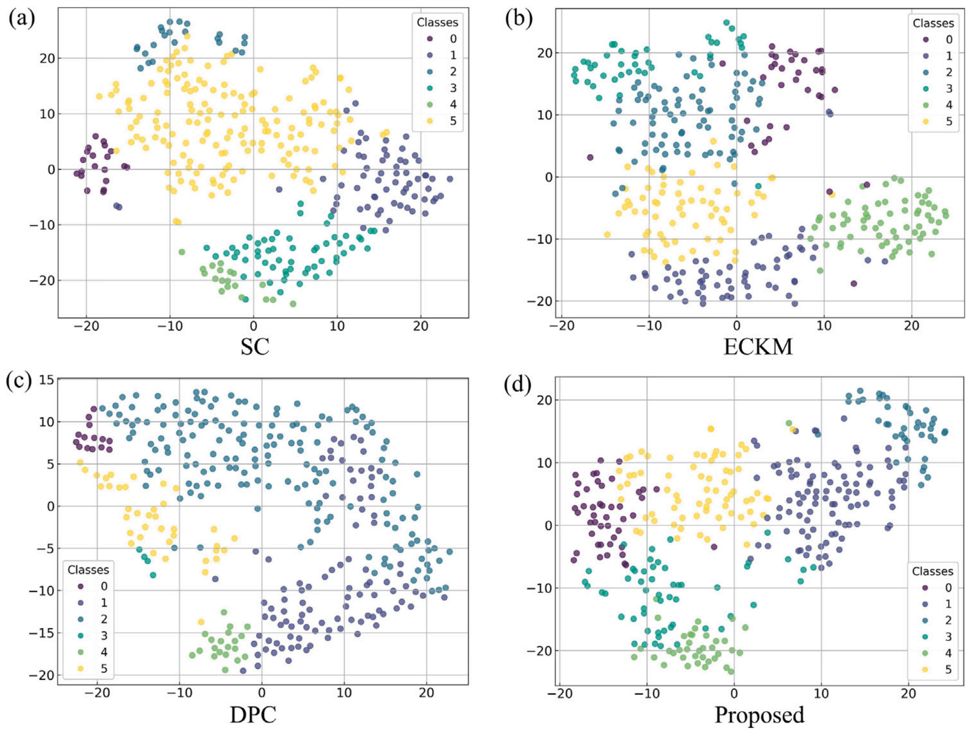

Fig. 3. t-SNE visualizations of four clustering algorithms in the 1st PV station.

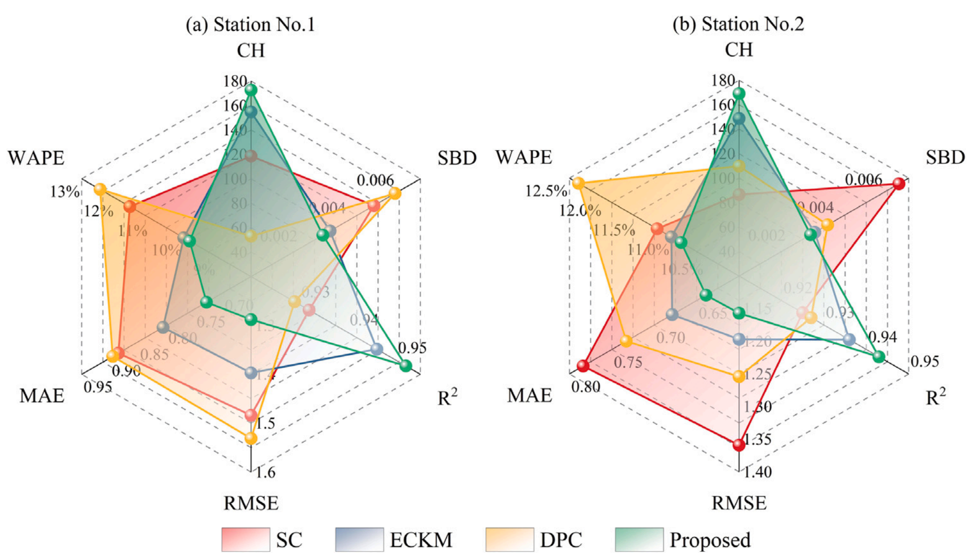

Fig. 4. Evaluation indicators of different clustering algorithms for two PV stations.

clustering approach that integrates computational geometry. It employs the Voronoi diagram and other methods to initialize centroids, thereby identifying optimal void regions to improve clustering performance. DPC (Rodriguez and Laio, 2014) is a clustering method based on density peaks, which partitions data points into clusters by identifying sample points with high local density and large distances as cluster centers.

To assess the ability of CWK-Medoids to enhance clustering accuracy, the clustering evaluation indices of various clustering algorithms for two PV power stations are calculated, as shown in Table 4. T-distributed stochastic neighbor embedding (t-SNE) is utilized to lower the dimensionality of the clustering results for visualization. The t-SNE visualizations of four clustering algorithms in the 1st PV station are illustrated in Fig. 3.

Table 4
Computed CH and SBD for different clustering algorithms.

Station No.	Model	SC	ECKM	DPC	Proposed
1	CH	118.555	154.763	53.006	172.672
	SBD	5.083E−03	3.269E−03	5.954E−03	2.971E−03
2	CH	86.594	148.669	109.763	169.018
	SBD	6.599E−03	3.126E−03	3.655E−03	2.953E−03

It can be observed that CWK-Medoids outperforms the other three clustering algorithms. Quantitatively, at station No. 1, the proposed model outperforms the worst-performing clustering approach (DPC), with an increase of 119.666 in CH and a decrease of 2.983E−03 in SBD; and surpasses the top-performing clustering approach (ECKM),

Table 5
Computed R^2, $RMSE$, MAE, and $WAPE$ for integrating different clustering algorithms and LagBoostNet of the 1st PV station.

Model	Scenario No.	0	1	2	3	4	5	AVG
SC	R^2	0.928	0.970	0.956	0.840	0.941	0.958	0.932
	$RMSE$	1.646	0.950	1.446	2.065	1.499	1.298	1.484
	MAE	1.121	0.572	0.848	1.252	0.822	0.696	0.885
	$WAPE$/%	13.387	5.741	11.765	19.272	9.925	9.384	11.579
ECKM	R^2	0.935	0.948	0.957	0.965	0.925	0.945	0.946
	$RMSE$	1.562	1.251	1.139	1.293	1.681	1.454	1.397
	MAE	0.938	0.725	0.601	0.654	1.08	0.839	0.806
	$WAPE$/%	11.254	7.812	10.169	8.927	12.898	8.858	9.986
DPC	R^2	0.960	0.920	0.946	–	0.897	0.924	0.929
	$RMSE$	1.180	1.634	1.446		1.971	1.426	1.531
	MAE	0.573	0.986	0.772		1.364	0.781	0.895
	$WAPE$/%	9.713	11.453	8.241		16.288	16.600	12.459
Proposed	R^2	0.979	0.944	0.958	0.944	0.938	0.948	0.952
	$RMSE$	0.785	1.297	1.227	1.459	1.527	1.434	1.288
	MAE	0.490	0.729	0.659	0.828	0.850	0.820	0.729
	$WAPE$/%	4.876	12.892	12.734	9.890	10.094	8.385	9.812

Note: "–" indicates that the sample size for the corresponding category is insufficient for regression modeling.

Table 6
Computed R^2, $RMSE$, MAE, and $WAPE$ for integrating different clustering algorithms and LagBoostNet of the 2nd PV station.

Model	Scenario No.	0	1	2	3	4	5	AVG
SC	R^2	0.901	–	0.931	0.946	0.923	–	0.925
	$RMSE$	1.478		1.350	1.177	1.432		1.359
	MAE	0.889		0.775	0.672	0.799		0.784
	$WAPE$/%	10.575		11.299	11.296	11.663		11.208
ECKM	R^2	0.921	0.963	0.919	0.922	0.938	0.955	0.936
	$RMSE$	1.453	1.017	1.340	1.446	0.923	1.001	1.197
	MAE	0.822	0.537	0.831	0.789	0.503	0.594	0.679
	$WAPE$/%	12.033	8.096	10.372	11.394	17.072	7.018	10.998
DPC	R^2	0.918	0.926	–	0.899	0.933	0.961	0.927
	$RMSE$	1.342	1.397		1.507	1.302	0.723	1.254
	MAE	0.798	0.770		0.930	0.729	0.436	0.733
	$WAPE$/%	13.212	11.177		11.154	12.875	13.416	12.367
Proposed	R^2	0.969	0.932	0.934	0.934	0.942	0.949	0.943
	$RMSE$	0.955	1.338	1.333	1.335	1.116	0.863	1.157
	MAE	0.540	0.747	0.742	0.745	0.596	0.463	0.639
	$WAPE$/%	6.292	10.893	10.820	10.860	12.301	13.961	10.855

Note: "–" indicates that the sample size for the corresponding category is insufficient for regression modeling.

with an increase of 17.909 in CH and a decrease of 2.980E−04 in SBD. At station No. 2, the proposed model outperforms the worst-performing clustering approach (SC), with an increase of 82.424 in CH and a decrease of 3.646E−03 in SBD; and surpasses the top-performing clustering approach (ECKM), with an increase of 20.349 in CH and a decrease of 1.730E−04 in SBD.

To verify the effectiveness of CWK-Medoids in improving prediction accuracy, this section integrates four clustering algorithms with the proposed LagBoostNet model, which is applied in day-ahead PPF. The prediction index values for the two PV power stations in different clustering scenarios, along with their average values, are compared, as shown in Tables 5–6. Here, "-" indicates that the sample size for the corresponding category is insufficient for regression modeling. Additionally, radar charts displaying six evaluation indicators for the two sites of the four aforementioned algorithms are presented in Fig. 4, where the predicted indicator values represent the average of six clustering scenarios. It is to present an overall performance comparison of each clustering algorithm at the different PV stations.

As shown in Tables 5–6: at station No. 1, the proposed model results in a 0.023 increase in the average R^2, while reducing the average $RMSE$, MAE, and $WAPE$ by 0.243, 0.166, and 2.647%, respectively, compared to the worst-performing clustering algorithm (DPC). In comparison to the most effective clustering approach (ECKM), the proposed approach shows a 0.006 increase in the average R^2, along with a decrease of 0.109, 0.077, and 0.174% in the average $RMSE$, MAE, and $WAPE$, respectively. At station No. 2, the proposed model results in a 0.018 increase in the average R^2, and a reduction of 0.202, 0.145, and 0.353% in the average $RMSE$, MAE, and $WAPE$, respectively, compared to the worst-performing clustering algorithm (SC). Notably, the algorithm with the worst $WAPE$ is not SC, but DPC, and the proposed algorithm reduces the $WAPE$ by 1.512% in comparison. When compared with the most effective clustering approach (ECKM), the proposed approach results in a 0.007 increase in the average R^2, and decreases the average $RMSE$, MAE, and $WAPE$ by 0.040, 0.040, and 0.143%, respectively. Additionally, for relatively underperforming algorithms (such as DPC at station No. 1 and SC and DPC at station No. 2), some individual clusters obtained from clustering have too few samples to establish regression models. In contrast, the proposed CWK-Medoids clustering algorithm did not exhibit this issue at either station. Each subset within its clustering results contained a sufficient number of samples to support independent training and evaluation of prediction models. This relatively balanced partitioning did not rely on manual intervention or dataset-specific optimization but benefited from the algorithm's inherent stability and robustness. However, we also recognize that in other datasets or application scenarios, particularly those involving rare situations such as rapidly changing or extreme weather, individual clusters might contain insufficient samples, potentially hindering the construction of predictive models. In such cases, strategies like merging clusters with similar meteorological features or introducing data augmentation techniques could be considered to enhance modeling capabilities. These strategies will be further explored in future research.

It is found that the performance of different clustering algorithms is generally proportional to each other in terms of clustering and prediction evaluation indices for two PV stations. Furthermore, the proposed CWK-Medoids performs optimally compared to the other three clustering algorithms across six evaluation indices.

4.3. Comparison of prediction models

The proposed clustering-ensemble learning model employs CWK-Medoids to partition the data into six clustering scenarios, and then LagBoostNet is utilized for PPF within each cluster. To ensure the comparative experiment is more reasonable and effective, the average forecasting accuracy of the proposed model across six scenarios (Proposed_AVG), along with the accuracy of LagBoostNet and eight prediction models, is evaluated in this section. To ensure the comprehensiveness and impartiality of the evaluation, the experiment selected comparative methods covering three categories of mainstream predictive model architectures, aiming to demonstrate the performance advantages of the proposed method in day-ahead PPF compared to mainstream approaches. Specifically, these include: (1) Transformer-based models: iTransformer and Informer; (2) RNN-based models: long short-term memory (LSTM), gated recurrent unit (GRU), and dual-stage attention-based recurrent neural network (DARNN); (3) CNN-based models: TimesNet, modern temporal convolutional network (ModernTCN), and fully convolutional networks (FCN). The core innovations for each comparative model are overviewed below: iTransformer (Liu et al., 2024) is an inverted Transformer architecture that utilizes the attention mechanism and the feedforward network on data with swapped dimensions. Informer (Zhou et al., 2021) efficiently processes long sequences by combining the self-attention mechanism with probabilistic sparsity. LSTM (Cao et al., 2023) and GRU (Cao et al., 2025) exhibit strong memory and long-term dependency capture capabilities when processing sequence data. LSTM effectively preserves and updates long-term information by incorporating cell states and gating mechanisms. The GRU structure is simpler than LSTM and offers better training speed and efficiency. DARNN (Yang et al., 2024) combines RNN with a two-stage attention mechanism to dynamically extract both the instantaneous fluctuations and time-dependent features of the input variable. TimesNet (Wu et al., 2023) represents a temporal sequence in the form of two-dimensional tensors, capturing both intra-cycle and inter-cycle changes while performing well in time series prediction, completion, classification, and anomaly detection tasks. ModernTCN (Luo and Wang, 2024) is an enhanced model based on the traditional temporal convolutional network (TCN) that captures both local and global patterns in the sequence through convolutional operations. FCN (Long et al., 2015) was initially introduced for image semantic segmentation. It substitutes the fully connected layer commonly found in conventional CNNs with a convolutional layer to achieve pixel-level semantic segmentation. This study extends these eight methods to regression prediction, establishes a real-time mapping relationship between NWP and measured PV data, and compares them with the proposed method.

Table 7 presents the effectiveness of ten models in two PV stations. To facilitate a clear comparison, a typical day from each clustering scenario of station No. 1 is extracted to represent the day-ahead PPF, and the prediction results are subsequently visualized. Each typical day contains 49 sample points, as illustrated in Fig. 5. Additionally, the comparison of prediction indicators for each model is presented in Fig. 6, providing an overall performance comparison of ten models in different stations.

It is evident that the proposed approach surpasses eight baseline methods. Quantitatively, at station No. 1, the effectiveness of the proposed approach shows an increase of 0.097 in R^2, and reductions of 1.055, 0.767, and 8.057% in $RMSE$, MAE, and $WAPE$, respectively, in contrast to the least performing approach (FCN). In comparison with the best-performing baseline method (iTransformer), the proposed model improves R^2 by 0.020 and reduces $RMSE$, MAE, and $WAPE$ by 0.320, 0.289, and 2.447%, respectively. At station No. 2, the performance of the proposed model shows an improvement of 0.081 in R^2, and reductions of 0.755, 0.749, and 9.379% in $RMSE$, MAE, and $WAPE$, respectively, in contrast to the least performing approach (FCN). In comparison with the best-performing baseline method (iTransformer), the proposed model improves R^2 by 0.017 and reduces $RMSE$, MAE, and $WAPE$ by 0.244, 0.143, and 0.613%, in that order. As shown in Table 8, the proposed clustering-ensemble learning model increases the average training time by approximately 7.5 min and the total inference time by about 0.2 s compared to iTransformer across both sites. Considering that the key requirement of PV forecasting systems lies in the response speed during real-time deployment, this additional computational cost remains within an acceptable range. Moreover, the proposed method achieves improvements of 0.020 and 0.017 in the R^2 metric compared to the best-performing baseline at the two sites, respectively. In the context of day-ahead PPF, where even slight improvements in accuracy can significantly impact operational scheduling or economic returns, such gains are of considerable value. Therefore, the proposed model strikes a reasonable balance between computational cost and forecasting performance, demonstrating its practical significance.

In contrast to LagBoostNet without clustering, the overall forecasting capability of the suggested model has also improved. Quantitatively, at station No. 1, the average values of R^2, $RMSE$, MAE, and $WAPE$ across the six clustering scenarios for the proposed model improve by 0.009 and decrease by 0.178, 0.112, and 0.229%, respectively, compared to using the LagBoostNet model alone. At station No. 2, these values improve by 0.012 and decrease by 0.193, 0.116, and 0.161%, respectively. When combining the content of this section with Tables 5–6, it can be noted that the overall forecasting capability is enhanced when the proposed CWK-Medoids and ECKM are integrated with LagBoostNet, while the performance is reduced when DPC and SC are combined with LagBoostNet. This suggests that reasonable clustering preprocessing before prediction can enhance the accuracy of day-ahead PPF, whereas unreasonable clustering preprocessing may have an adverse effect.

Furthermore, the proposed ensemble learning model LagBoostNet demonstrates significant predictive accuracy advantages over eight baseline models. At station No. 1, LagBoostNet achieves an R^2 of 0.943 with $RMSE$, MAE, and $WAPE$ values of 1.466, 0.841, and 10.041% respectively, representing an R^2 improvement of 0.011 and reductions

Table 7
Computed R^2, $RMSE$, MAE, and $WAPE$ for different forecasting models.

Station No.	Model	R^2	RMSE	MAE	WAPE/%
1	Timesnet	0.912	1.823	1.250	15.239
	ModernTCN	0.882	2.103	1.356	16.075
	iTransformer	0.932	1.608	1.018	12.259
	informer	0.922	1.719	1.144	13.356
	DARNN	0.929	1.639	1.081	12.909
	LSTM	0.922	1.721	1.199	14.326
	GRU	0.920	1.738	1.226	14.646
	FCN	0.855	2.343	1.496	17.869
	LagBoostNet	0.943	1.466	0.841	10.041
	Proposed_AVG	0.952	1.288	0.729	9.812
2	Timesnet	0.913	1.521	1.043	15.892
	ModernTCN	0.869	1.854	1.184	17.131
	iTransformer	0.926	1.401	0.782	11.468
	informer	0.923	1.423	0.887	13.271
	DARNN	0.925	1.409	0.864	12.596
	LSTM	0.919	1.464	0.946	13.803
	GRU	0.919	1.464	0.952	13.884
	FCN	0.862	1.912	1.388	20.234
	LagBoostNet	0.931	1.350	0.755	11.016
	Proposed_AVG	0.943	1.157	0.639	10.855

Table 8
Computational efficiency comparison of models.

Model	Station No.	Training time (min)	Inference time (s/day)
iTransformer	1	17.1	0.633
	2	16.7	0.571
Proposed	1	24.7	0.832
	2	24.2	0.814

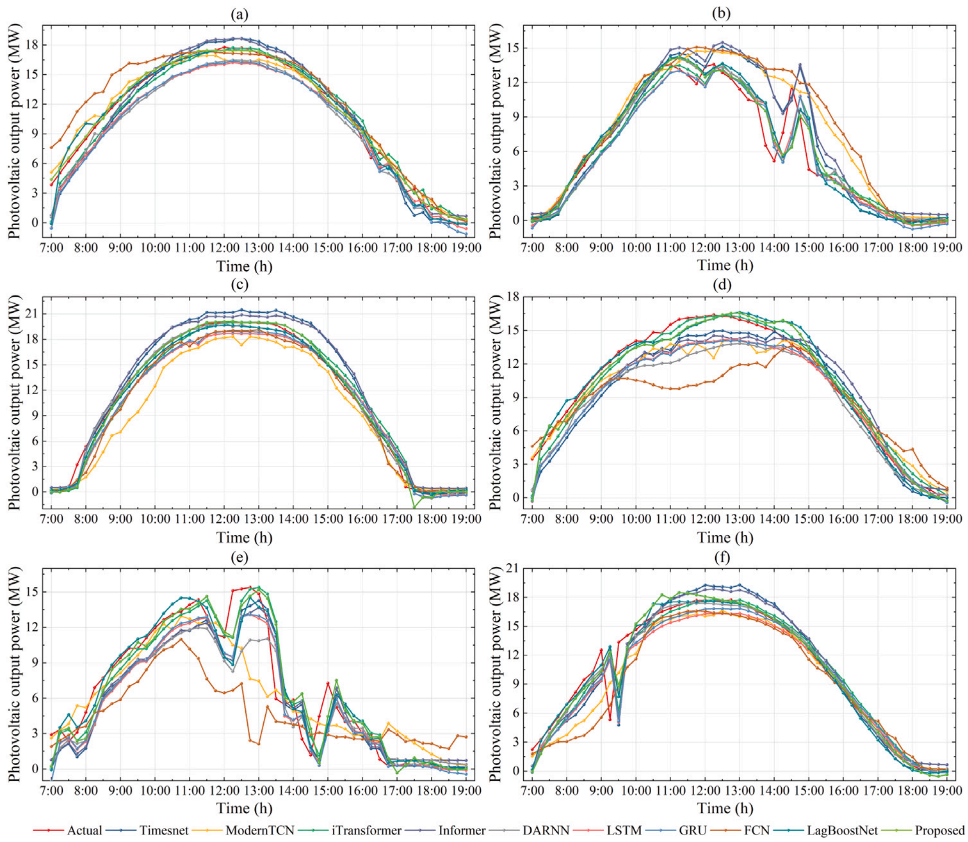

Fig. 5. Comparison of predicted and actual values for a typical day in multiple scenarios of the 1st PV station.

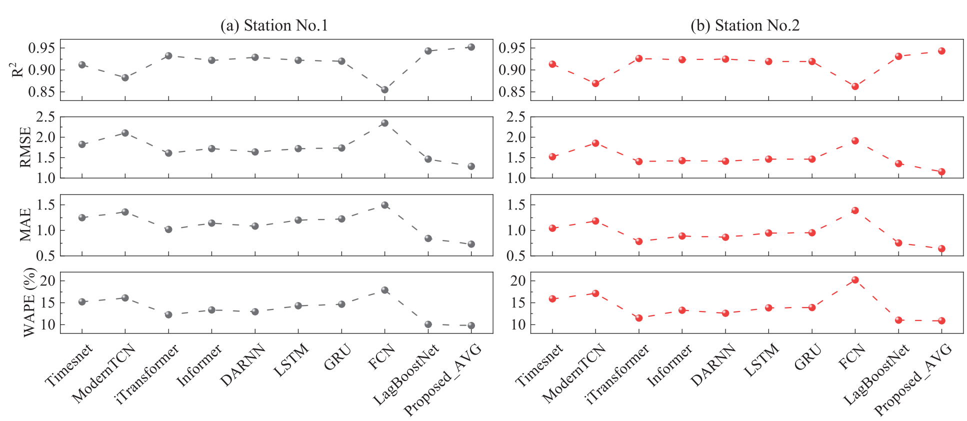

Fig. 6. Comparison of prediction indicators.

of 0.142 in $RMSE$, 0.177 in MAE, and 2.218% in $WAPE$ compared to the best-performing baseline model iTransformer. At Station No. 2, it attains an R^2 of 0.931 with $RMSE$, MAE, and $WAPE$ values of 1.350, 0.755, and 11.016%, corresponding to an R^2 gain of 0.005 and decreases of 0.051 in $RMSE$, 0.027 in MAE, and 0.452% in $WAPE$ against iTransformer. These enhancements primarily originate from the strategic ensembling of CatBoost and LightGBM coupled with the integration of lag features, which collectively strengthen the model's capacity to handle nonlinear high-dimensional temporal data.

4.4. Model interpretation and ablation experiment

We propose a novel deep time-series clustering algorithm, CWK-Medoids, combined with an ensemble learning model based on lag features, LagBoostNet, for day-ahead PPF. To assess the performance of each structure of CWK-Medoids, it is compared with three constructed clustered ablation models: WK-Medoids, CK-Medoids, and K-Medoids. These are the clustering models obtained by removing CAE, the DTW distance, and both CAE and the DTW distance from CWK-Medoids, respectively. The validation of each structure of CWK-Medoids under the clustering and predictive evaluation index system is presented in Tables 9–11. Table 9 presents the metrics comparison of the individual clustering models. Tables 10–11 present the comparison of the predictive metrics for the fused clustering ablation model and the proposed LagBoostNet model.

As shown in Table 9: at station No. 1, the CH of CWK-Medoids is 2.346 higher and the SBD is 5.000E−05 lower than that of WK-Medoids; it is 14.937 higher and 2.640E−04 lower than that of CK-Medoids; and 25.047 higher and 3.250E−04 lower than that of K-Medoids, respectively. At station No. 2, the CH of CWK-Medoids is 11.646 higher

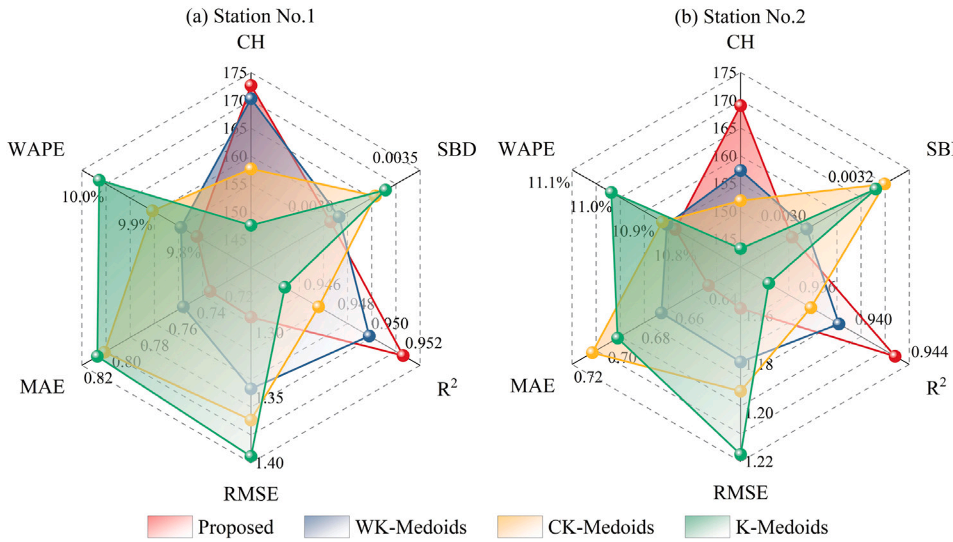

Fig. 7. Evaluation indicators of clustering ablation models for two PV stations.

Table 9
Clustering evaluation indicators of clustering ablation models.

Station No.	Model	Proposed	WK-Medoids	CK-Medoids	K-Medoids
1	CH	172.672	170.326	157.735	147.625
	SBD	2.971E−03	3.021E−03	3.235E−03	3.296E−03
2	CH	169.018	157.372	151.949	143.318
	SBD	2.953E−03	2.996E−03	3.227E−03	3.201E−03

and the SBD is 4.300E−05 lower than that of WK-Medoids; it is 17.069 higher and 2.740E−04 lower than that of CK-Medoids; and 25.700 higher and 2.480E−04 lower than that of K-Medoids, respectively.

As shown in Tables 10–11, the fusion of CWK-Medoids exhibits a higher average R^2 and lower average RMSE, MAE, and WAPE when compared to the fusion of clustering ablation models. At station No. 1, CWK-Medoids improves R^2 by 0.002 and reduces $RMSE$, MAE, and $WAPE$ by 0.055, 0.019, and 0.033%, respectively, in contrast to WK-Medoids; by 0.005, 0.079, 0.075, and 0.092%, respectively, compared to CK-Medoids; and by 0.007, 0.107, 0.080, and 0.202%, respectively, compared to K-Medoids. At station No. 2, the improvements are 0.004 for R^2 and reductions of 0.022, 0.028, and 0.020% for $RMSE$, MAE, and $WAPE$ compared to WK-Medoids; 0.006, 0.034, 0.069, and 0.031% compared to CK-Medoids; and 0.009, 0.060, 0.054, and 0.152% compared to K-Medoids. The radar chart, illustrating the six evaluation indicators of the four models discussed above at two PV stations, is presented in Fig. 7, where the predicted evaluation indicator values are the average values under the six clustering scenarios.

To assess the performance of each structure in LagBoostNet, a comparison is made with the constructed ablation model, BoostNet, as shown in Table 12. BoostNet is the ensemble model obtained by removing the lag features from LagBoostNet. It can be observed that at station No. 1, the R^2 of LagBoostNet increases by 0.008, while $RMSE$, MAE, and $WAPE$ decrease by 0.100, 0.109, and 1.291%, respectively, compared to BoostNet. At station No. 2, R^2 increases by 0.005, while $RMSE$, MAE, and $WAPE$ showing reductions of 0.048, 0.020, and 0.362%, respectively.

The proposed method exhibits high interpretability, primarily manifested through:

(1) At the prediction framework level, the clustering-ensemble learning model demonstrates higher overall prediction accuracy compared to its non-clustering counterpart. This suggests that appropriate clustering preprocessing prior to prediction can effectively enhance the accuracy of day-ahead PPF. This may be attributed to the fact that targeted modeling for each clustering scenario allows adaptation to data non-uniformity. It also facilitates better capture of PV system operating modes under varying meteorological conditions and more accurate data fitting within each cluster.

(2) At the deep time-series clustering algorithm level, CWK-Medoids demonstrates superior performance across three ablation models in both clustering and prediction metrics, validating the efficacy of its CAE and DTW distance components. Combined with prior comparative experiments, these results confirm CWK-Medoids' optimal performance over other clustering algorithms for weather pattern clustering and day-ahead PPF when integrated with LagBoostNet, likely attributable to: (i) The deep clustering algorithm extracts useful features from high-dimensional NWP data, overcoming traditional clustering algorithms' limitations in computational load and performance with high-dimensional data. CAE is particularly suitable for sequence data with spatial or temporal dependencies. When used in clustering tasks, its convolutional layers extract local temporal patterns, effectively capturing time-dependent properties of weather factors. (ii) The constructed WWED can measure similarity from both temporal and spatial dimensions, considering the trend homogeneity and spatial similarity of the daily NWP series, while overcoming the shortcomings of using only the Euclidean distance for time-series feature capturing.

(3) At the ensemble learning model level, LagBoostNet surpasses BoostNet, highlighting the benefits of incorporating lag features. Together with previous comparative experiments, these results demonstrate that LagBoostNet achieves optimal performance compared to the eight baseline approaches. This implies that the ensemble learning model based on lag features is more suitable for day-ahead PPF, likely because: (i) The proposed model integrates CatBoost and LightGBM, combining the strengths of both in handling nonlinear features and high-dimensional data while mitigating the instability associated with the reliance on individual models. (ii) The inclusion of lag features facilitates the capture of potential patterns in PV historical data over time, compensating for the inadequacy of existing ensemble models that only consider weather dependence and ignore temporal patterns.

4.5. Sensitivity analysis

To evaluate the generalization performance of the proposed method, a sensitivity analysis is conducted in this section. Specifically, independent Gaussian noise is added to all input features (excluding timestamps and target variables) in both datasets to simulate potential data perturbations that may occur in real-world environments. The noise follows a normal distribution with a mean of 0 and a standard

Table 10
Prediction evaluation indicators of integrating clustering ablation models and LagBoostNet (the 1st PV station).

Model	Scenario No.	0	1	2	3	4	5	AVG
Proposed	R^2	0.979	0.944	0.958	0.944	0.938	0.948	0.952
	RMSE	0.785	1.297	1.227	1.459	1.527	1.434	1.288
	MAE	0.490	0.729	0.659	0.828	0.850	0.820	0.729
	WAPE/%	4.876	12.892	12.734	9.890	10.094	8.385	9.812
WK-Medoids	R^2	0.948	0.943	0.948	0.959	0.940	0.959	0.950
	RMSE	1.602	1.463	1.407	1.137	1.340	1.110	1.343
	MAE	0.792	0.840	0.799	0.610	0.806	0.643	0.748
	WAPE/%	9.853	10.035	8.387	10.242	13.621	6.930	9.845
CK-Medoids	R^2	0.945	0.932	0.981	0.948	0.938	0.938	0.947
	RMSE	1.448	1.578	0.774	1.415	1.269	1.719	1.367
	MAE	0.826	0.876	0.491	0.840	0.719	1.069	0.804
	WAPE/%	9.862	9.243	4.837	8.941	15.863	10.676	9.904
K-Medoids	R^2	0.912	0.938	0.957	0.958	0.964	0.943	0.945
	RMSE	1.826	1.525	1.141	1.125	1.303	1.448	1.395
	MAE	1.190	0.844	0.607	0.653	0.692	0.869	0.809
	WAPE/%	14.216	10.016	10.260	7.153	9.232	9.208	10.014

Table 11
Prediction evaluation indicators of integrating clustering ablation models and LagBoostNet (the 2nd PV station).

Model	Scenario No.	0	1	2	3	4	5	AVG
Proposed	R^2	0.969	0.932	0.934	0.934	0.942	0.949	0.943
	RMSE	0.955	1.338	1.333	1.335	1.116	0.863	1.157
	MAE	0.540	0.747	0.742	0.745	0.596	0.463	0.639
	WAPE/%	6.292	10.893	10.820	10.860	12.301	13.961	10.855
WK-Medoids	R^2	0.926	0.968	0.920	0.938	0.953	0.932	0.939
	RMSE	1.398	0.956	1.405	1.076	0.898	1.343	1.179
	MAE	0.771	0.561	0.826	0.617	0.479	0.748	0.667
	WAPE/%	11.210	7.244	11.461	10.002	14.422	10.913	10.875
CK-Medoids	R^2	0.945	0.923	0.945	0.932	0.961	0.918	0.937
	RMSE	1.094	1.412	0.927	1.337	1.052	1.326	1.191
	MAE	0.707	0.771	0.500	0.747	0.613	0.908	0.708
	WAPE/%	8.460	11.331	15.439	11.008	7.264	11.812	10.886
K-Medoids	R^2	0.916	0.945	0.936	0.910	0.940	0.956	0.934
	RMSE	1.362	1.203	1.306	1.519	0.912	1.002	1.217
	MAE	0.864	0.619	0.663	0.885	0.493	0.633	0.693
	WAPE/%	10.749	9.932	10.553	12.117	15.128	7.562	11.007

Table 12
Computed R^2, $RMSE$, MAE, and $WAPE$ for ablation models of LagBoostNet.

Station No.	Model	R^2	*RMSE*	*MAE*	*WAPE*/%
1	LagBoostNet	0.943	1.466	0.841	10.041
	BoostNet	0.935	1.566	0.950	11.332
2	LagBoostNet	0.931	1.350	0.755	11.016
	BoostNet	0.926	1.398	0.775	11.378

deviation equal to 5% of the absolute value of the original features. For variables with physical constraints, boundary condition processing is applied, such as ensuring non-negative variables do not take negative values and wind direction values remain within 0° to 360°. The model's predictive performance is evaluated on the perturbed dataset and compared with evaluation metrics from the original dataset, with results presented in Table 13. The listed metric values represent the average performance across six clustering scenarios. The relative change percentage (Change/%) is calculated based on the following formula: $\Delta M = (M_{\text{Noisy}} - M_{\text{Original}})/M_{\text{Original}} \times 100\%$.

Experimental results demonstrate that the proposed clustering-ensemble learning model maintains superior prediction accuracy even with 5% Gaussian noise perturbation. Compared to the original data, all evaluation metrics exhibit relative changes within ±2.5%, indicating minor variations and excellent model robustness. Taking Site 1 as an example, the R^2 slightly decreases by 0.315%, while the $RMSE$, MAE, and $WAPE$ increase by 1.553%, 1.097%, and 0.805%, respectively; however, the overall prediction accuracy remains at a high level. In comparison, Site 2 exhibits slightly larger metric fluctuations, but the overall performance remains within a stable range, further demonstrating the generalization capability of the proposed method across different scenarios.

Table 13
Prediction performance comparison on original and 5%-noisy data.

Station No.	Metric	Original	Noisy (5%)	Change/%
1	R^2	0.952	0.949	−0.315
	$RMSE$	1.288	1.308	+1.553
	MAE	0.729	0.737	+1.097
	$WAPE$/%	9.812	9.891	+0.805
2	R^2	0.943	0.938	−0.530
	$RMSE$	1.157	1.181	+2.074
	MAE	0.639	0.649	+1.565
	$WAPE$/%	10.855	10.954	+0.912

5. Conclusions

To address the insufficient weather pattern recognition capability in day-ahead PPF and inadequate focus on temporal patterns in existing multivariate regression models, this study proposes a novel prediction framework integrating the deep time-series clustering algorithm CWK-Medoids with an enhanced ensemble learning model LagBoostNet. Experiments were conducted using NWP data and measured PV output power from two stations in Hebei, China. Through comparative experiments, ablation studies, and sensitivity analysis, the framework's predictive performance and generalization capability were systematically evaluated. Key findings include:

(1) The proposed clustering-ensemble learning model outperforms multiple benchmarks, achieving R^2 of 0.952 and 0.943 on two datasets

— improvements of 0.020 and 0.017 over the best baseline. Compared to standalone LagBoostNet (without clustering), it shows R^2 gains of 0.009 and 0.012.

(2) The CWK-Medoids algorithm significantly surpasses other clustering methods in weather pattern partitioning and day-ahead forecasting when combined with LagBoostNet, yielding R^2 improvements of 0.006 and 0.007 over the top alternative clustering algorithm.

(3) LagBoostNet alone demonstrates superior forecasting performance, exceeding the best baseline by 0.011 and 0.005 in R^2 on two datasets.

In summary, the proposed model exhibits high accuracy, generalizability, and interpretability for day-ahead PPF. Interpretability stems from: (i) the clustering-ensemble learning framework's adaptability to data heterogeneity; (ii) CWK-Medoids' capability for refined weather pattern segmentation via CAE integration and WWED design; and (iii) LagBoostNet's improved modeling of nonlinear, high-dimensional temporal data through ensemble tree learning and lagged feature incorporation.

This research offers practical value for improving operational dispatch and grid PV integration. Future work will explore the model's adaptability across multi-regional/climatic contexts and enhance real-time forecasting via model compression techniques.

CRediT authorship contribution statement

Feifei Yang: Writing – original draft, Software, Methodology. **Xueqian Fu:** Writing – review & editing, Funding acquisition, Conceptualization. **Zhengshuo Li:** Funding acquisition. **Dawei Qiu:** Supervision. **Hamed Badihi:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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